- linear combinations
- if you multiply by 0

System of Linear Equations
Inverse
Determinants

matrix: rectangular

- Diagonal matrix
- Symmetric matrix
  - symmetric around the diagonal.
- Transpose
  - flips rows with columns.
- Multiply matrix with vector → vector.

CS 556
Erisa Terolli

Lec2

System of Linear Equations

Matrix-Vector Multiplication

Three vectors $\mathbf{u} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \mathbf{w} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \in \mathbb{R}^3$.

The linear combinations in $\mathbb{R}^3$ are $x_1\mathbf{u} + x_2\mathbf{v} + x_3\mathbf{w}$.

$$x_1 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$Ax = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 - x_1 \\ x_3 - x_2 \end{bmatrix}$$

Matrix times vector
Linear combination of columns in the matrix.

$$Ax = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = b$$

Which combinations of *u, v, w* produces a particular vector *b*?

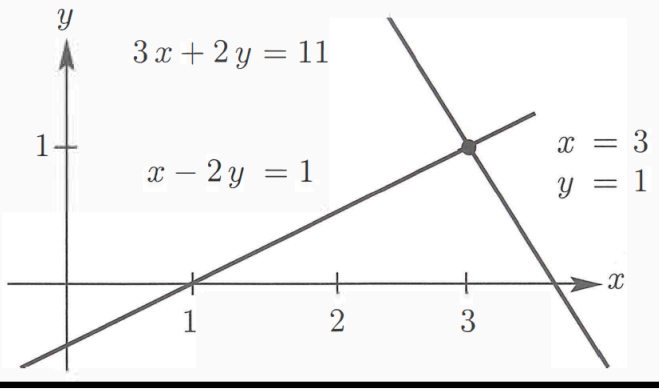
Solving Linear Equations

$$x - 2y = 1$$
$$3x + 2y = 11$$

Column Picture

$$\mathbf{x}\begin{bmatrix}1\\3\end{bmatrix} + \mathbf{y}\begin{bmatrix}-2\\2\end{bmatrix} = \begin{bmatrix}1\\11\end{bmatrix}$$

Row Picture



Matrix Equation

$$Ax = b$$

$$\begin{bmatrix}1 & -2\\3 & 2\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}1\\11\end{bmatrix}$$

3 equations in 3 unknowns

$$x + 2y + 3z = 6$$
$$2x + 5y + 2z = 4$$
$$6x - 3y + z = 2$$

Row Picture

The row picture shows three planes meeting at a single point

Column Picture

$$\mathbf{x}\begin{bmatrix}1\\2\\6\end{bmatrix} + \mathbf{y}\begin{bmatrix}2\\5\\-3\end{bmatrix} + \mathbf{z}\begin{bmatrix}3\\2\\1\end{bmatrix} = \begin{bmatrix}6\\4\\2\end{bmatrix}$$

Matrix Equation

$$Ax = b$$

$$\begin{bmatrix}1 & 2 & 3\\2 & 5 & 2\\6 & -3 & 1\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}6\\4\\2\end{bmatrix}$$

Ax comes from dot products, each row times the column *x*.

$$\mathbf{Ax} = \begin{bmatrix}(row\ 1)\cdot x\\(row\ 2)\cdot x\\(row\ 3)\cdot x\end{bmatrix}$$

Ax is a combination of column vectors.

$$Ax = x(col\ 1) + y(col\ 2) + z(col\ 3)$$

21 − 15 = 6

15 − 6

Systems of Linear Equations SLE

① $x + y + 3z = 15$

$x + 2y + 4z = 21$

$x + y + 2z = 13$

② $x + y = 3 \rightarrow x = 3 - y$

$x - y = 1 \quad x - y = 1$

③ Transformations allowed

1. Row exchange
2. Multiply a row with a #
3. Add 2 rows together.

④ How do we SIMPLIFY! the system

⑤ For each row, find the 1st nonzero value

↳ convert the rest to 0

pivot

$$\begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 21 \\ 13 \end{bmatrix}$$

REF → RREF

$$\begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ -2 \end{bmatrix}$$

⑥

$$\begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ 2 \end{bmatrix}$$

$R1 = R1 - R2$

⑦

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 6 \\ 2 \end{bmatrix}$$

$R1 = R1 - 2R3$

$R2 = R2 - R3$

⑧

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \\ 2 \end{bmatrix}$$

How to solve SLE?

$$\begin{bmatrix} 1 & 1 & 3 \\ 1 & 2 & 4 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 21 \\ 13 \end{bmatrix}$$

Key to solving SLE these elementary transformation, that keep the solution set the same, but that transform the equation system into a simpler form:

- Exchange of two rows
- Multiplication of a row with a constant
- Addition of two rows

Gaussian Elimination

Performs elementary transformation to bring a system of linear equation into reduced row-echelon form.

A matrix is in row-echelon form if:

- All rows that contain only zeros are at the bottom of the matrix
- Looking at nonzero rows only, the first nonzero number (aka pivot) is always strictly to the right of the pivot above it.

An matrix is in reduced row echelon form if:

- It is in row-echelon form
- Every pivot is 1
- The pivot is the only nonzero entry in its column

$$\begin{bmatrix} 1 & 3 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 3 \\ 1 & 2 & 4 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 21 \\ 13 \end{bmatrix}$$

$$\begin{array}{ll} & \begin{bmatrix} 1 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 21 \\ 13 \end{bmatrix} \\ \text{R2 - R1} & \begin{bmatrix} 1 & 2 & 4 \end{bmatrix} \\ \text{R3 - R1} & \begin{bmatrix} 1 & 1 & 2 \end{bmatrix} \end{array}$$

$$\begin{array}{ll} & \begin{bmatrix} 1 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 21 \\ 13 \end{bmatrix} \\ \text{R2 - R1} & \begin{bmatrix} 1 & 2 & 4 \end{bmatrix} \\ \text{R3 - R1} & \begin{bmatrix} 1 & 1 & 2 \end{bmatrix} \end{array} \qquad \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ -2 \end{bmatrix}$$

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$$\begin{array}{l} \\ \\ \text{-R3} \end{array} \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ -2 \end{bmatrix}$$

$$\begin{array}{ll} & \begin{bmatrix} 1 & 1 & 3 \\ 1 & 2 & 4 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 21 \\ 13 \end{bmatrix} \\ \text{R2 - R1} & \\ \text{R3 - R1} & \end{array} \qquad \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ -2 \end{bmatrix}$$

$$\begin{array}{ll} & \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ -2 \end{bmatrix} \\ \text{-R3} & \end{array} \qquad \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ 2 \end{bmatrix}$$

$$\begin{matrix} \\ \text{R2 - R1} \\ \text{R3 - R1} \end{matrix} \begin{bmatrix} 1 & 1 & 3 \\ 1 & 2 & 4 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 21 \\ 13 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 1 & 3 \\ 1 & 2 & 4 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 21 \\ 13 \end{bmatrix}$$

R2 - R1

R3 - R1

$$\begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ 2 \end{bmatrix}$$

-R3

$$\begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ 2 \end{bmatrix}$$

R1 - 3R3

R2 - R3

$$\begin{array}{l} \\ \text{R2 - R1} \\ \text{R3 - R1} \end{array} \begin{bmatrix} 1 & 1 & 3 \\ 1 & 2 & 4 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 21 \\ 13 \end{bmatrix} \qquad \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ -2 \end{bmatrix}$$

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$$\begin{matrix} \\ \text{R2 - R1} \\ \text{R3 - R1} \end{matrix}\begin{bmatrix} 1 & 1 & 3 \\ 1 & 2 & 4 \\ 1 & 1 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 21 \\ 13 \end{bmatrix} \qquad \begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ -2 \end{bmatrix}$$

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-R3

$$\begin{matrix} \text{R1 - 3R3} \\ \text{R2 - R3} \\ \\ \end{matrix}\begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ 2 \end{bmatrix} \qquad \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 4 \\ 2 \end{bmatrix}$$

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$$\begin{matrix} \text{R1 - 3R3} \\ \text{R2 - R3} \\ \\ \end{matrix}\begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ 2 \end{bmatrix} \qquad \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 4 \\ 2 \end{bmatrix}$$

$$\begin{matrix} \text{R1 - R2} \\ \\ \\ \end{matrix}\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 4 \\ 2 \end{bmatrix}$$

R2 - R1

R3 - R1

$$\begin{bmatrix} 1 & 1 & 3 \\ 1 & 2 & 4 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 21 \\ 13 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ -2 \end{bmatrix}$$

-R3

$$\begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ 2 \end{bmatrix}$$

R1 - 3R3

R2 - R3

$$\begin{bmatrix} 1 & 1 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 4 \\ 2 \end{bmatrix}$$

R1 - R2

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 4 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \\ 2 \end{bmatrix}$$

$$\begin{matrix}\\ \text{R2 - R1}\\ \text{R3 - R1}\end{matrix}\begin{bmatrix}1 & 1 & 3\\ 1 & 2 & 4\\ 1 & 1 & 2\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}15\\ 21\\ 13\end{bmatrix}$$

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$$\begin{matrix}\\ \\ \text{-R3}\end{matrix}\begin{bmatrix}1 & 1 & 3\\ 0 & 1 & 1\\ 0 & 0 & -1\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}15\\ 6\\ -2\end{bmatrix}$$

$$\begin{bmatrix}1 & 1 & 3\\ 0 & 1 & 1\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}15\\ 6\\ 2\end{bmatrix}$$

$$\begin{matrix}\text{R1 - 3R3}\\ \text{R2 - R3}\\ \\ \end{matrix}\begin{bmatrix}1 & 1 & 3\\ 0 & 1 & 1\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}15\\ 6\\ 2\end{bmatrix}$$

$$\begin{bmatrix}1 & 1 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}9\\ 4\\ 2\end{bmatrix}$$

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$$\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}5\\ 4\\ 2\end{bmatrix}$$

REDUCED ROW-ECHELON FORM

Determinant

Determinant

- The determinant of a square matrix $A \in \mathbb{R}^{nxn}$ is a function that maps A into a real number.

- Notation $det(A) = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$

- Only for square matrix

Basic Properties of Determinants

- The determinant of the identity matrix is 1.
- The determinant changes sign when two rows are exchanged.
- The determinant is a linear function of each row separately.

$$\begin{bmatrix} ta & tb \\ c & d \end{bmatrix} = t\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\begin{bmatrix} a+a' & b+b' \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} a' & b' \\ c & d \end{bmatrix}$$

Derived Properties of Determinants

how can we actually compute?

- If two rows of matrix A are equal, then det(A) = 0.
- Subtracting a multiple of one row from another row leaves the determinant of matrix A unchanged.
- A matrix with a row of zeros has det(A) = 0.
- If A is triangular then det(A) is the product of diagonal entries.
- If A is singular, then det(A) = 0. If A is invertible, then det(A) ≠ 0.
- $det(AB) = det(A)det(B)$
- $det(A^T) = det(A)$

Laplace Expansion

① Fix 1 row & expand with respect to that row / column.

Consider a matrix $A \in \mathbb{R}^{nxn}$. Then for all $j = 1,\ldots,n$:

1. Expansion along column j

$$det(A) = \sum_{k=1}^{n} (-1)^{k+j}\, a_{kj}\, det(A_{k,j})\,.$$

2. Expansion along row j

$$det(A) = \sum_{k=1}^{n} (-1)^{k+j}\, a_{jk}\, det(A_{j,k})\,.$$

Here $A \in \mathbb{R}^{(n-1)x(n-1)}$ is the sub matrix of A
that we can obtain when deleting row k and column j.

Example

$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

(1)

$\det(A) = 1 \cdot \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} + (-1)(2) \cdot \begin{vmatrix} 3 & 2 \\ 0 & 1 \end{vmatrix} + 3 \begin{vmatrix} 3 & 1 \\ 0 & 0 \end{vmatrix}$

(2) $= 1 \cdot 1 - 2 \cdot 3 + 0$

$= 1 - 6$

$= -5$ $\therefore$ this matrix has an inverse.

Example

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$det(A) = (-1)^{1+1} \cdot 1 \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix}$$

$$+(-1)^{1+2} \cdot 2 \begin{vmatrix} 3 & 2 \\ 0 & 1 \end{vmatrix}$$

$$+(-1)^{1+3} \cdot 3 \begin{vmatrix} 3 & 1 \\ 0 & 0 \end{vmatrix}$$

$$= 1(1-0) - 2(3-0) + 3(0-0) = -5$$

Determinant Applications

Geometry

- Area/Volume of shape specified by coordinates in the matrix

Matrix Inverse

- Divide by determinant. No inverse of matrix if determinant is zero

Matrix Inverse

geometrically

- undoes a linear transformation.
- tells you whether a transformation preserves enough information to be undone.

Inverse Matrix

Consider a square matrix $A \in \mathbb{R}^{nxn}$. Let matrix $B \in \mathbb{R}^{nxn}$ have the property that $AB = I_n = BA$.

B is called the inverse of A and denoted by A^{-1}.

Not every matrix A possesses an inverse A^{-1}.

If this inverse exists, matrix A is called regular/invertible/ nonsingular, otherwise singular/noninvertible.

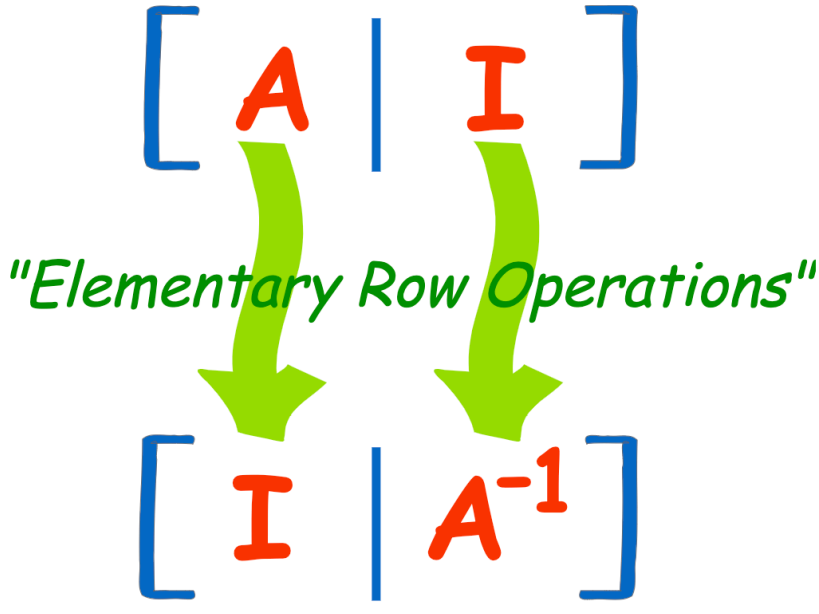
$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 1 \\ 4 & 4 & 5 \\ 6 & 7 & 7 \end{bmatrix} \qquad \mathbf{A}^{-1} = \begin{bmatrix} -7 & -7 & 6 \\ 2 & 1 & -1 \\ 4 & 5 & -4 \end{bmatrix}$$

Invertibility Test

- The algorithm to test invertibility is elimination: Matrix A_{nxn} must have n(non-zero) pivots.
- The algebra test for invertibility is the determinant of A : $det(A)$ must be non zero.
- The equation that test for invertibility is $Ax = 0$: where $x = 0$ must be the only solution.

if the only solth is the 0 vector,

Gauss-Jordan Elimination



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Cofactors Method

- The **minors matrix**: a matrix of determinants
- The **cofactors matrix**: the minors matrix element-wise multiplied by a grid of alternating +1 and -1. runs in factorial time...
- The **adjugate matrix**: the transpose of the cofactors matrix
- The **inverse matrix**: the adjugate matrix divided by the determinant

Inverse for a 2x2 Matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$A^{-1} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix} = \frac{1}{det(A)} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$$

Inverse for a 3x3 Matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 1 \\ 4 & 4 & 5 \\ 6 & 7 & 7 \end{bmatrix} \qquad \mathbf{Minors} = \begin{bmatrix} -7 & -2 & +4 \\ +7 & +1 & -5 \\ +6 & +1 & -4 \end{bmatrix}$$

$$\mathbf{Cofactors} = \begin{bmatrix} -7 & +2 & +4 \\ -7 & +1 & +5 \\ +6 & -1 & -4 \end{bmatrix}$$

$$\mathbf{A}^{-1} = \begin{bmatrix} -7 & -7 & 6 \\ 2 & 1 & -1 \\ 4 & 5 & -4 \end{bmatrix}$$

$$\mathbf{Adjugate} = \begin{bmatrix} -7 & -7 & +6 \\ +2 & +1 & -1 \\ +4 & +5 & -4 \end{bmatrix}$$